Project Report: Industrial Predictive Maintenance System

Project Title: Predictive Maintenance using Linear Regression

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1. Executive Summary & Problem Statement

The Problem: In industrial manufacturing, "Reactive Maintenance" (repairing machines only after they fail) leads to significant financial losses, unscheduled downtime, and safety hazards.

The Goal: This project aims to build an analytical tool that monitors sensor data and predicts the probability of failure before it occurs, allowing for "Proactive Maintenance."

2. Technical Concepts

This project applies several core Data Science and AI foundations:

- **Intelligent Agents:** The system acts as a diagnostic agent that perceives its environment through sensors (Temperature, Torque) and provides a risk-based output.
- **Statistical Foundations:** Statistical distributions were used to analyze sensor noise and identify trends in data degradation.
- **Supervised Learning:** A **Linear Regression** model was implemented to map continuous sensor inputs to a failure risk score.
- **Feature Engineering:** High-dimensional data from multiple sensors was analyzed to determine which mechanical factors (like Tool Wear) most significantly impact machine health.

3. Methodology & Approach

1. **Data Acquisition:** Utilized the **AI4I 2020 Predictive Maintenance Dataset**, containing 10,000 records of industrial sensor data.
2. **Exploratory Data Analysis (EDA):** Created scatter plots and histograms to identify correlations between high process temperatures and mechanical failures.
3. **Data Preprocessing:** Cleaned the dataset by removing non-numeric identifiers and splitting the data into a **Training Set (80%)** and a **Test Set (20%)**.
4. **Model Training:** Developed a Regression model to find the mathematical relationship between sensor readings and the likelihood of failure.
5. **Inference Implementation:** Developed an **Interactive Prediction Feature** that allows for real-time manual data entry for instant risk assessment.

4. Key Engineering Decisions

- **Regression vs. Classification:** I chose to use **Regression** because it provides a "Risk Percentage" (0-100%) rather than a simple "Yes/No." This allows maintenance teams to prioritize machines with 90% risk over those with 50% risk.
- **Primary Features:** I prioritized **Torque** and **Tool Wear** as the primary indicators, as these variables showed the strongest statistical correlation with mechanical failure during the testing phase.

5. Challenges Faced

- **Imbalanced Data:** In real-world industrial settings, failures are much rarer than healthy states. I had to ensure the model could accurately flag these rare events without producing excessive false alarms.
- **Feature Scaling:** Managing variables with vastly different ranges (e.g., Rotational Speed in thousands vs. Torque in tens) required careful data handling to ensure all features contributed appropriately to the prediction.

6. Key Learnings

This project provided hands-on experience in the complete Machine Learning lifecycle:

- Moving from raw sensor data to actionable business intelligence.
- Understanding how environmental factors (Temperature) and mechanical stress (Torque) interact to affect equipment longevity.
- Designing an interactive interface to make complex AI predictions accessible to non-technical users.

7. Conclusion

The developed system demonstrates that predictive modeling can significantly improve industrial safety and efficiency. Future iterations could involve moving toward **Deep Learning** or **Anomaly Detection** algorithms to identify even subtler patterns of machine degradation before they lead to failure.